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Fostering socio-emotional competencies in children on the autism spectrum using a parent-assisted serious game: A multicenter randomized controlled trial

Simone Kirst[a,*], Robert Diehm[b], Katharina Bögl[a], Sabine Wilde-Etzold[c], Christiane Bach[d], Michele Noterdaeme[c], Luise Poustka[e], Matthias Ziegler[f], Isabel Dziobek[a]

[a] Clinical Psychology of Social Interaction, Berlin School of Mind and Brain, Department of Psychology, Humboldt-Universität zu Berlin, Berlin, Germany
[b] Department of Child and Adolescent Medicine, Medical University of Vienna, Vienna, Austria
[c] Department of Child and Adolescent Psychiatry and Psychotherapy, Josefinum Augsburg, Augsburg, Germany
[d] Department of Psychosomatic Medicine and Psychotherapy, University of Leipzig, Leipzig, Germany
[e] Department of Child and Adolescent Psychiatry and Psychotherapy, University Medicine, Georg-August Universität Göttingen, Göttingen, Germany
[f] Psychological Diagnostics, Department of Psychology, Humboldt-Universität zu Berlin, Berlin, Germany

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ABSTRACT

Objective: Serious games are a promising means of fostering socio-emotional skills in children on the autism spectrum (AS). However, empathy and related constructs have not yet been addressed comprehensively and together with emotion recognition, and there is a lack of randomized controlled trials (RCT) to investigate skill maintenance and the transfer to functional behavior.

Method: The manualized, parent-assisted serious game *Zirkus Empathico* (ZE) was tested against an active control group, in a six-week multicenter RCT. Eighty-two children aged 5–10 years on the AS were assessed at baseline, post-treatment, and three-month follow-up. Empathy and emotion recognition skills were defined as the primary outcomes. The secondary outcomes included measures of emotional awareness, emotion regulation, autism social symptomatology (Social Responsiveness Scale), and subjective therapy goals.

Results: Training effects were observed after the intervention for empathy ($d = 0.71$) and emotion recognition ($d = 0.50$), but not at follow-up. Moderate effects on emotional awareness, emotion regulation, and autism social symptomatology were indicated by the short and mid-term assessments. Parents reported treatment goal attainment and positive training transfer.

Conclusion: While a six-week training with ZE failed to induce lasting changes in empathy and emotion recognition, it may be effective for improving emotional awareness and emotion regulation, and mitigate general autism symptomatology.

Clinical trial registration information: Zirkus Empathico – Promoting socioemotional competencies in 5- to 10-year-old children with autism spectrum conditions using a computer-based training program; https://www.drks.de/; DRKS-ID: DRKS00009337; Universal Trial Number (UTN): U1111-1175-5451.

Abbreviations: AS, autism spectrum; RCT, randomized controlled trial; ZE, Zirkus Empathico.

* Corresponding author.
E-mail addresses: simone.kirst@hu-berlin.de (S. Kirst), robert.diehm@meduniwien.ac.at (R. Diehm), katharina.boegl@hu-berlin.de (K. Bögl), WildeE.Sabine@josefinum.de (S. Wilde-Etzold), christiane.bach@medizin.uni-leipzig.de (C. Bach), Noterdaeme.Michele@josefinum.de (M. Noterdaeme), luise.poustka@med.uni-goettingen.de (L. Poustka), zieglema@hu-berlin.de (M. Ziegler), isabel-dziobek@hu-berlin.de (I. Dziobek).

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1. Introduction

Understanding one's own emotional state and that of others enables us to interact effectively, thus supporting prosocial behavior (Decety & Meyer, 2008). Persons on the autism spectrum (AS)[1] have been shown to exhibit differences in different facets of empathy (Song, Nie, Shi, Zhao, & Yang, 2019), which might result in divergent reactions to other people's emotions, peer conflicts, and social exclusion (Hill & Frith, 2003). As an umbrella term (Hodges & Myers, 2007), empathy comprises emotional resonance to others' emotions (Decety, 2011), which arises from the comprehension of others' emotional states through perspective-taking (*cognitive empathy;* Baron-Cohen & Wheelwright, 2004), but also bottom-up processes (Decety, 2011) such as mirroring mechanisms (Bird & Viding, 2014). Emotional resonance might either trigger *personal distress,* a self-oriented motivation to reduce stress by withdrawal, or *empathic concern*, a motivation to care for or to help someone in need (Decety & Lamm, 2009).

As suggested by neurobiological models of empathy (e.g., Bird & Viding, 2014; Shamay-Tsoory, 2011), recognizing others' emotions (e.g., through facial expressions) represents leverage to empathic understanding and sharing others' emotions. Indeed, emotion recognition was found to be positively related to empathic concern, but negatively to personal distress (Israelashvili, Sauter, & Fischer, 2020). Furthermore, emotion recognition (e.g., Israelashvili, Oosterwijk, Sauter, & Fischer, 2019), as well as cognitive and affective facets of empathy (e.g., Bird et al., 2010; Moriguchi et al., 2006; Moriguchi et al., 2007), are associated with differentiating own emotions.

As recently reported in a meta-analysis by Song et al. (2019), a large body of research points to difficulties in cognitive empathy and reduced empathic concern in persons on the AS. On the contrary, emotional resonance seems to be intact or even enhanced (Dziobek et al., 2008; Smith, 2009; Song et al., 2019), which is resulting in higher personal distress compared to non-autistic individuals, partly due to problems in emotion regulation in emotion regulation (goal-directed monitoring and modification of emotional responses; Eisenberg & Spinrad, 2004; Mazefsky et al., 2013). In turn, individuals on the AS might aim to reduce their distress by withdrawal when confronted with feelings of another person, instead of attending to the other person's emotional state, which possibly results in inadequate social behavior (Decety & Lamm, 2009). Beyond that, persons on the AS are reported to show poor recognition of facial emotions (see meta-analyses by Lozier, Vanmeter, & Marsh, 2014; Uljarevic & Hamilton, 2013) and difficulties in describing their feelings and inner states (e.g., Giannotti, Falco, & Venuti, 2020; Griffin, Lombardo, & Auyeung, 2016; Milosavljevic et al., 2016). Taken together, research of the past decades has generated evidence for deviations in empathy, which are characterized by difficulties in cognitive empathy, enhanced personal distress, reduced empathic concern, and difficulties in underlying competencies (e.g., emotion recognition, emotional awareness, and emotion regulation), which have a negative impact on prosocial behavior.

In recent years, computerized intervention programs, also known as *serious games*, have been implemented to train social-cognitive skills in children on the AS. Serious games offer a cost-effective solution to bridge the gap between the high need for evidence-based interventions and limited access to specialist autism services (Casale et al., 2015). By fulfilling the desire for structure and predictability while reducing stress through limiting social demands and meeting the interest in technology, serious games are a fruitful complement to face-to-face interventions for children on the AS (Bölte, Golan, Goodwin, & Zwaigenbaum, 2010). Indeed, several studies addressing social-cognitive skills have reported enhanced attention and motivation, as well as positive training results (see Kouo & Egel, 2016; Mazon, Fage, & Sauzéon, 2019, for recent reviews). An RCT, which was conducted in the US, investigated the effectiveness of the 12-weeks program "*Mind Reading*" (Baron-Cohen, Golan, Wheelwright, & Hill, 2004) in 43 children on the AS aged 7–12 years with normal intelligence (IQ > 85). The study found significantly better emotion decoding and encoding skills (primary outcomes) in the treatment group post-treatment, which were maintained at a five-week follow-up (Thomeer et al., 2015). Tanaka et al. (2010) reported improvements in facial recognition and processing skills in 42 children, adolescents, and young adults on the AS after training 20 h with the "*Let's Face It!*" program relative to a matched control group. For the interactive program "*FaceSay*™", RCTs by Hopkins et al. (2011) and Rice, Wall, Fogel, and Shic (2015) showed improvements in affect recognition, mentalizing, and social skills in two samples (n = 49; n = 31) of primary-school-aged children on the AS after 6–10 weeks of training. A cross-cultural evaluation in the UK, Israel, and Sweden showed, that 6 to 9-year-old participants on the AS with normal intelligence using the serious game "*Emotiplay*" for 8–12 weeks showed significant improvement on facial, vocal, body, and integrative emotion recognition tasks (Fridenson-Hayo et al., 2017). Finally, studies investigating the effectiveness of the Australian serious game (SAS, Beaumont & Sofronoff, 2008) in home- and school-based settings, found treatment gains in social skills, emotion regulation, and child anxiety reduction after 7–8 weeks of treatment, and at follow-up (Beaumont & Sofronoff, 2008; Einfeld et al., 2018; Sofronoff, Silva, & Beaumont, 2017).

However, most of these serious games have been tested and made available in English-speaking countries only and very few/none integrated persons on the autism spectrum into the design process. Further, they focused predominantly on emotion recognition and mentalizing, while enhanced personal distress and reduced emotional concern (Song et al., 2019) in response to others' feelings have rarely been targeted. Since aberrant social behavior might result from an interplay of cognitive and affective facets of empathy and deficits in underlying competencies (e.g., emotion recognition, emotional awareness), this might partly explain the finding that previous evaluations of serious games showed little evidence of transfer effects to real-world settings (Grynszpan, Weiss, Perez-Diaz, & Gal, 2014).

Concerning methodology, although randomized controlled trials (RCTs) have become more common in recent years (Fletcher-Watson, 2014), few studies on the effectiveness of serious games have used follow-up assessment to assess maintenance and generalization effects (Berggren et al., 2017) or used performance tests of social skills (Mazon et al., 2019). Furthermore, only a few studies (Hopkins et al., 2011; Rice et al., 2015; Young & Posselt, 2012) incorporated active control groups to account for the amount of time spent playing on the computer and social interactions with the training facilitators during training. However, these studies did not address maintenance effects.

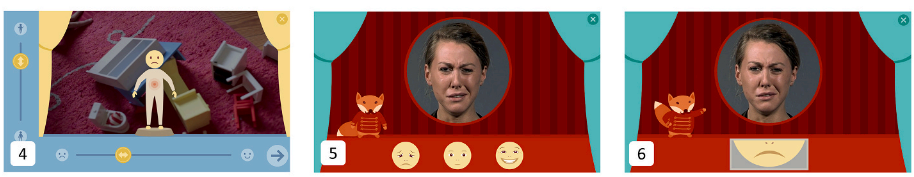
To conclude, we aimed to apply a more holistic approach targeting empathy and related constructs here with our newly developed serious game *Zirkus Empathico* (ZE) for children on the AS. To foster empathic behavior and emotion understanding in real-world settings, ZE focuses in its first modules on foundations of empathy, thus, awareness and differentiation of own emotional states (module I; compare SAS, Beaumont & Sofronoff, 2008) and emotion recognition from facial expressions (module II). Next, cognitive empathy is addressed by teaching inferring others' emotions from emotion eliciting contexts (module III; compare *Emotiplay*; Fridenson-Hayo et al., 2017). Finally, ZE targets dealing with personal distress through conveying that others' emotions can elicit own emotions and through teaching possible options of prosocial and empathic behavior to given contexts (module IV).

While the comparable SAS program (Beaumont & Sofronoff, 2008) was designed to suit the cognitive profile and needs of children aged from 8 to 12 years old with normal intelligence (IQ > 85), ZE addresses younger children of between 5 and 10 years, with varying intellectual

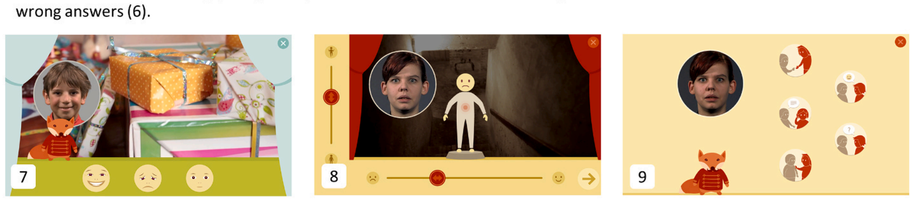
[1] To respect the language preferences of the autism community, we use the term "persons on the autism spectrum", which was voted as being least offensive and most accepted by the community in recent publications (e.g., Bury, Jellett, Spoor, & Hedley, 2020; Kenny et al., 2016; Bottema-Beutel, Kapp, Lester, Sasson, & Hand, 2021).

level from low to high (IQ > 70), by using simplified language and age-appropriate, visualized content (e.g., animated emotional states, Fig. 1). To meet the actual needs of the children and their families, the development of the serious game contained co-design elements. Finally, we integrated naturalistic video-based content and allowed caregivers to assist with the training at home to enhance maintenance and generalization (compare Sofronoff et al., 2017). We conducted a multicenter RCT in Germany and Austria to test our primary hypothesis (H1) that a six-week training course would result in improvements in emotion recognition and empathy at the end of the training (T2) and at a three-month follow–up (T3). Specifically, we expected empathy, as measured by a parent questionnaire tapping cognitive and affective aspects of empathy, and emotion recognition, which was assessed by a behavioral paradigm, to be enhanced in the ZE training group at T2 and T3. Training effects should be expressed by differences between the groups' developmental trajectories over time (H1.1) and group differences at post-treatment and follow-up with effect sizes favoring the ZE training group (TG) (H1.2). In addition, we included several secondary outcome measures to quantify effects on emotional awareness and clinically relevant functions and traits, which have been shown to be related to understanding own and others' emotions such as emotion regulation (Kashdan, Barrett, & McKnight, 2015) and callous-unemotional traits (Leno et al., 2015). Further, we explored the transfer of emotion recognition skills to untrained stimulus material, real-life social behavior, i.e. autism social symptomatology, and well-being. Finally, the moderating effects of autism symptomatology, verbal age, and nonverbal IQ were investigated, and the relationships between changes in emotional awareness/emotion regulation over the course of the training and changes in emotion recognition/empathy three months after the intervention were explored. It was hoped that the

Emotion Library (1,2): The library contains an explanation of the emotion manikin and the 6 emotion cards with definitions and explanations of the basic emotions/neutral states targeted. (1): Characteristic display of emotion in the eye/mouth region, synonyms for the emotion label, examples of emotion triggering contexts, arousal and valence of the emotion indicated by the manikin (2). **Emotion manikin/Transfer module (3):** The interactive manikin is a central element in the game which enables one's own inner emotional states and those of others to be visualized. The child/tutor can use two five-step sliders to indicate a) his/her level of bodily arousal (very calm – calm – neutral – aroused – highly aroused), and b) the valence of his/her feeling (very negative – negative – indifferent/neutral – positive – very positive). The manikin changes its movement and facial expression accordingly.

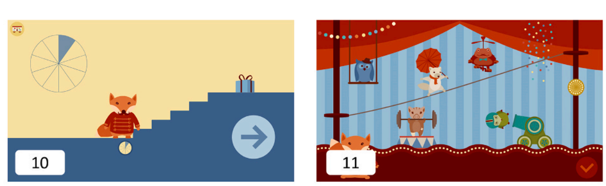


Module I – Awareness of own emotions (4): By using the emotion manikin, the child specifies its inner emotional state regarding a specific context which is presented using a short emotion-inducing video clip. In a second step, the child describes its inner state by choosing an emotion label. The program provides feedback on whether the inner state and chosen emotion label are matched correctly or incorrectly. **Module II – Emotion recognition in faces (5,6):** The child is asked to identify the correct emotion label for the emotion shown by the facial expression (5). Unsolved tasks are repeated, with correct answers being prompted by visual hints after the first 2 wrong answers and an automatic correction after 3 wrong answers (6).



Module III – Cognitive empathy (7): The module requires the child to identify a specific emotion-eliciting context, in which another person finds him/herself. The child chooses the correct emotion label from 3 options (5). Unsolved tasks are repeated, with correct answers being prompted by verbal hints and a picture of the other person's emotional expression in response to the context (6). **Module IV – Empathy and prosocial action (8,9):** The stimuli show a third person's emotional expression embedded in the emotion triggering context. By using the emotion manikin, the child first evaluates its own emotional state, which may be provoked by the other person's emotional reaction to the given context (8). In a second step (9), the child is asked to choose between (a) approaching the person, (b) leaving the situation, or (c) waiting and seeing. If "approach" is selected, it is presented with a selection of concrete prosocial actions towards the other person (e.g. helping, being friendly, comforting, listening).

Level System (10): The system indicates the child's progress within each module. The modules are comprised of 6 (Mod. I, III, IV) or 12 levels (Mod. II). Each level contains 10 tasks, which are displayed as a pie chart which is completed with the number of solved tasks. When a level has been completed, the child is allowed to choose an animated reward, which is integrated into a circus arena. Extra rewards are hidden in boxes to enhance motivation.



Reward System (11): In total, 30 animated reward elements (e.g. bouncing ball, spinning top) and 5 animated animals can be added to the circus environment with increasing performance.

Fig. 1. Modules and elements of the serious game *Zirkus Empathico*.

results would provide important implications for the potential of serious games in teaching socio-emotional skills to children on the AS.

2. Method

2.1. Study design

A six-week multicenter, parallel-group, randomized, pragmatic clinical trial (DRKS-ID: DRKS00009337; Universal Trial Number (UTN): U1111-1175-5451) was conducted at Humboldt–Universität zu Berlin, Germany (HU) and two child and adolescent psychiatry and psychotherapy university departments with specialized outpatient clinics for children/adolescents on the AS in Augsburg (KJPP AUG), Germany, and Vienna (MedUni Wien), Austria. 82 children aged 5–10 years on the AS participated in the trial between December 2015 and April 2018. The study was registered in the German register for clinical studies and received ethical approval from the Ethics Committee at HU (2015/10/07) and the clinical authorities of the two outpatient clinics. The registered study protocol was fulfilled except for one outcome measure being excluded from analysis because the majority of the children did not comprehend task requirements (see below). The coordinating activities, data management, and analysis were conducted at HU. In accordance with the CONSORT 2010 Checklist (Schulz, Altman, & Moher, 2010), detailed descriptions of the sample selection, settings, training protocol, and outcome measures are available online (see Table S1 in online supplement).

2.2. Study participants

Children (5 years–10 years, 11 months) with a clinical consensus ICD-10 (WHO, 1994) diagnosis of childhood autism, Asperger syndrome, atypical autism, or pervasive developmental disorder not otherwise specified (PSS–NOS), were eligible for the study. Diagnosis was established by specialized and experienced multi-professional teams (e.g., child and adolescents psychiatrists, neuropediatrics, psychologists, and other therapists like speech and occupational therapists, rehabilitation educators), using a variety of measures (e.g., parent interviews, patient observation, clinical interviews, basic functional level assessments) and clinical judgment (compare NICE guidelines, German guidelines of DGKJP; see Table S2 for diagnostic institutions). For seventy-three participants, the study operators have been provided with results of the Autism Diagnostic Observation Schedule (ADOS-G/ADOS-2; Lord et al., 2000; Lord et al., 2015) by caregivers or clinicians. Before the trial, the Autism Diagnostic Interview-Revised (ADI-R) short version (Hoffmann, Heinzel-Gutenbrunner, Becker, & Kamp-Becker, 2015) was administered to 80 participants to confirm the autism diagnosis. Autism symptomatology was further assessed by using the Social Communication Questionnaire (SCQ; Rutter, Bailey, & Lord, 2006) for 80 participants (see Table S3 for more detailed information on diagnostic scores). Autism symptomatology was further assessed by using the Social Communication Questionnaire (SCQ; Rutter et al., 2006) for 80 participants. Children with a nonverbal IQ below 70, as measured via the Colored Progressive Matrices (CPM) intelligence test (Raven, 2002) and an insufficient receptive German language level (verbal age$<$5,0; tested by Peabody Picture Vocabulary Test, PPVT, 4th revision (Dunn & Dunn, 2015) or the examiner's judgment) were excluded. Interfering neurological/medical conditions (except well–treated epilepsy), and very severe aggressive behavior were excluded by parental reporting. Further exclusion criteria were social skills training or intensive training in emotional competencies during the three months prior to or in parallel with the intervention, and the parallel participation in other clinical studies, according to parental report. The children in the HU sample were recruited through autism care units, parent organizations, and the study-website (www.zirkus-empathico.de). In the KJPP AUG, eligible children were asked to participate after receiving an autism diagnosis. The same procedure was used at the MedUni Wien, with additional children being recruited through parent organizations and autism care units. Written informed consent was obtained from the children's legal guardians. The children were tested in four sessions (diagnostics, T1/T2/T3) at their respective study centers, care unit, or at home. The families received 7 €/hour as compensation.

2.3. Intervention

The manualized, tablet-based ZE intervention (Kirst, Zoerner, Schütze, Lucke, & Dziobek, 2015; Zoerner, Schütze, Kirst, Dziobek, & Lucke, 2016; Fig. 1, demo video, and further description in Tables S1 and S3, online) was conceptualized by the first and the last author (HU) and developed by a team of independent designers and developers under the supervision of the first author and Potsdam University, Germany. ZE includes four modules that focus on (I) awareness of own emotions, (II) emotion recognition in faces, (III) inferring emotions from emotion-eliciting contexts (cognitive empathy), and (IV) understanding emotional resonance and learning appropriate reactions towards other people's emotions. A fifth module (interactive animation: emotion manikin) is used for emotional communication in family life (real-life transfer). The modules consist of different levels and open up according to the child's progress within the previous module (see Fig. S2, online). Open modules can be selected freely to allow a choice to be provided (compare Whyte, Smyth, & Scherf, 2015). The training is self-explanatory, with a fox character that guides the participants through the training modules by providing instructions, explanations, prompts, and rewards. It is recommended that the training is conducted under a caregiver's guidance to enhance emotional and empathic communication. The training includes video stimuli showing i) 315 adults and children's emotional facial expressions and ii) 62 emotion-eliciting situations filmed in the first-person perspective with an illustrated verbal introduction. Each video addresses a basic emotion (fear, anger, sadness, surprise, joy) or a neutral state. The production of video-taped adult expressions was part of a comprehensive project using 60 actors to produce an ecologically valid set of 40 emotions (Kliemann, Rosenblau, Bölte, Heekeren, & Dziobek, 2013). In addition, the videos of children's basic emotional expressions and context video stimuli were produced for ZE with help from the HU media services. All the stimuli were validated using expert ratings and showed high average emotion recognition rates and good believability (see S3a). During the serious game development process, children (n = 15), their parents, and adults on the AS (n = 2) repeatedly provided feedback, usability was tested, and the intervention was piloted (see S3b). None of these individuals took part in the RCT.

Training took place in the children's homes with tablets provided by the study centers. One caregiver was instructed to assist each child's training by discussing its content, providing motivation during training, and, importantly, supporting the transfer of skills to everyday life by using module V. The caregiver's involvement was intended to enable emotional communication within the family's natural home context. The caregivers received on-site instruction and a training manual (see Table S1 for details) and were supervised by training operators during the training period.

In total, the intervention lasted for six weeks with a minimum intensity of 100 min of training per week, to be done in a minimum of two single training sessions. Modules II/III were complemented by the task of analyzing emotional situations in children's movies (e.g. *Pippi Longstocking*) using the generalization module (40 min/week). In addition, the aim of transfer to contexts outside the ZE environment was addressed by requiring that module V be used for a minimum of 10 min per day for individualized, real-life transfer goals focused on recognizing one's own and other people's emotions and prosocial/empathic acts (e.g., playing a game recognizing emotional expressions in family members, for further examples, see Table S1, online). The transfer was supported by an additional, paper-based reward system (Table S1). The tablets were returned to the study centers at T2 to prevent nonmonitored training extension.

2.4. Control intervention

The active control training was framed as aiming to foster the children's confidence in their actions/knowledge. It was conducted as parent-assisted computerized training to guarantee a comparable level of motivation, media use, and quantity/quality of parent interaction. The caregivers providing the training received different serious games, which targeted non-social skills/knowledge. Depending on the children's age and interest, the serious games either focused on traffic safety (N = 29), body-related knowledge (N = 3), or school- and nature-related knowledge (N = 8; see Table S1 for additional information). Additionally, caregivers received a reward system for transfer goals, a training manual, and weekly supervision via phone. As in the ZE group, the caregivers were encouraged to teach the training content to their children actively, by using the respective app (e.g., talking about targeted behaviors, serving as role models, ensuring understanding) for 100-min per week in the context of a minimum of two sessions per week plus 10 min daily transfer.

2.5. Intervention monitoring: training intensity, treatment fidelity, and motivation

To ensure treatment fidelity, the training was monitored: (1) The caregivers of both groups recorded the training times using a paper-based system, and a tracking application was used to record training times automatically for ZE. (2) In both groups, the caregivers' commitment to comply with the study conditions was monitored by weekly phone calls from training operators who asked about the progress of training, training transfer, interfering events, and problems (see Table S1). The conversations were protocolled, and the caregivers received weekly emails with a summary of the conversation. After the intervention (T2), treatment fidelity was further assessed using a treatment satisfaction parental report (see below). (3) The children's motivation to engage in the training was assessed at five time-points during the course of the intervention. The assisting caregivers were asked to rate their child's behavior during gameplay on nine items (e.g., "explores novelty", "stays engaged") by using a four-point scale ranging from engaging in passive (1) to spontaneous (4) behavior. The items were adapted from the *Pediatric Volitional Questionnaire* (PVQ, Basu, Kafkes, Schatz, Kiraly, & Kielhofner, 2008), an observational assessment tool for children with developmental disorders (note: the instrument was not validated in a parent-sample). The children's motivation and enjoyment were also assessed in the treatment satisfaction report at T2.

2.6. Randomization and blinding

Eligible participants were randomized to either the TG or the active control group (CG). Minimization was performed with a randomization ratio of 0.8 and the two three-staged stratification factors verbal age and study center using MinimPy (Saghaei, 2011). Since ZE is an online psychotherapy with a clear focus on socio-emotional skills, blinding children and caretakers was not possible. In addition, for feasibility reasons at the respective sites, only one assistant on each site was responsible for all study-related tasks (e.g. training supervision, recruitment, testing), and thus, blinding of testers was not possible either.

2.7. Outcome measures

2.7.1. Primary outcomes

The primary outcomes were changes in the *Griffith Empathy Measure* (GEM, Dadds et al., 2008) total score for parent-rated empathy comprising affective and cognitive aspects. The GEM consists of 23 items for assessing empathy in children using a nine-point-rating scale. Reliability and validity were sufficient (Cronbach's α = 0.91, test-retest reliability over 6 months: r = 0.69). In addition, the *Kids Emotion Recognition Multiple Images Task* (KERMIT) served as the primary measure of changes in emotion recognition accuracy (Drimalla & Dziobek, 2019). In this task, 48 naturalistic pictures of adults' basic emotions with varying intensities are presented on a computer. Children must choose the correct emotion label from two options as quickly as possible. Analyses of 73 of the children in the total sample with valid data at baseline plus 64 additionally measured, non-autistic children, revealed that the *KERMIT* had good reliability (McDonald's Omega = 0.91). The test-retest reliability of the KERMIT was not assessed.

2.7.2. Secondary outcomes

Since the *KERMIT* pictures are frames extracted from some of the ZE training video clips, emotion recognition was additionally tested using 28 computer-presented photos of six basic emotions taken from the pictures in a facial affect set (Ekman & Friesen, 1976) to account for more distant generalization effects. The internal consistency in our sample at baseline (N = 73 children on the AS, N = 64 non-autistic children; compare KERMIT) was good (McDonald's Omega = 0.94). Changes in the awareness of one's own emotions were assessed using a modified version of the *Level of Emotional Awareness Scale for Children* (LEAS-C; Bajar, Ciarrochi, Lane, & Deane, Frank, 2005). The children reported their emotional reactions to 12 narratively presented, emotion-eliciting situations. The internal consistency of the LEAS-C subscale referring to own emotions (self-LEAS-C) was sufficient (Cronbach's α. = 0.71). Emotion regulation was tested with the emotion regulation (ERC-ER) and lability/negativity (ERC-LN) subscale of the *Emotion Regulation Checklist* parent questionnaire (ERC; Shields & Cicchetti, 1997). The internal consistency of both scales is adequate (ERC-L/N: α = 0.96; ERC-ER: α = 0.83). Callous-unemotional traits were measured by the *Inventory of Callous-Unemotional Traits* (ICU; Essau, Sasagawa, & Frick, 2006), the subscales of which showed acceptable reliability (Cronbach's α = 0.77 - 0.81; test–retest reliability over M = 23 days: r = 0.84; Moore et al., 2017). To further explore the training's ecological validity, the parents and (kindergarten) teachers/assistants rated changes in general autism social symptomatology using the *Social Responsiveness Scale* (SRS; Constantino & Gruber, 2007). The SRS has demonstrated excellent internal consistency (Cronbach's α = 0.97) and a good test-retest reliability of r = 0.77 (for females) to r = 0.85 (for males).The *Kiddy Kindl* (Ravens-Sieberer & Bullinger, 1998) parent questionnaire was used to measure the children's wellbeing. The internal consistency for the whole scale was acceptable, with Cronbach's α = 0.85. The *Multifaceted Empathy Test for Adolescents/Children* (MET-J/K; Dziobek et al., 2008; Poustka et al., 2010) was excluded from the analyses because the majority of the children did not demonstrate a full understanding of the instructions and the concept of the emotional facets of empathy (e.g., as shown by systematic or repetitive response tendencies; see also Table S1, online).

The assessments took place at baseline (T1), post-treatment (T2), and at the three-month follow-up (T3). T2 was conducted within the first two weeks after the end of the intervention. All the parent/teacher questionnaires were presented online using the SoSci Survey (Leiner, 2014)/Limesurvey platforms (Limesurvey GmbH; MedUni Wien).

2.7.3. Treatment goal achievement and treatment satisfaction

At baseline, we operationalized two individually defined, socio-emotional treatment goals for each child in the TG, using *Goal Attainment Scaling* (GAS; McDougall & King, 2007). At T2, parents were asked to rate their child's level of achievement on a scale from −2 'no change', through 0 'goal achieved', to 2 'change far beyond goal'. Finally, an unstandardized treatment satisfaction questionnaire was used post-treatment in the TG and CG, to target treatment satisfaction, acceptance, feasibility, treatment fidelity, and changes in generalized behavior that had not otherwise been assessed. Where possible, the children were also interviewed using a modified version of the questionnaire.

2.8. Statistical analyses

The sample size was planned on the medium effect sizes found in previous serious game interventions targeting emotion recognition (Grynszpan et al., 2014). A total sample of 82 participants was needed to provide 80% power (1- β) at a two-sided 5% α level and 7% attrition. Data from all randomized participants was used for primary and secondary analyses according to the intention-to-treat principle. The training effects were estimated using path analyses with maximum likelihood estimation and robust 'Huber-White' standard errors (MLR). This statistical approach was chosen because it combines several advantages: First, missing data can be handled by using a full information maximum likelihood estimation method (FIML; Enders & Bandalos, 2001). In contrast, classical MANOVA handles missing data by deleting all the individuals with incomplete data (Hox, 2000), resulting in a reduced sample size and lower statistical power. Second, it is possible to control for differences between groups at baseline, as well as in confounding variables, such as autism scores. Third, autoregressive approaches are easy to implement. In the context of our study, this means the following: Using path analyses enabled us to keep all the data points collected and adjust our baseline measurements for differences in baseline scores and autism symptomatology (SCQ). These adjusted values were then used as predictors for the subsequent time points, thus leading to less biased estimates for T2 and T3. All the variables were centered to zero (with 0 being the mean and positive scores indicating values above average) to facilitate the interpretation of intercepts.

We specified three different path models to test our two primary hypotheses: First, training effects should be expressed by differences in the two groups' developmental trajectories over time (H1.1). Statistically, this is indicated by different path coefficients between groups, with higher values for regression coefficients indicating higher stability between measured constructs over time, while lower values indicate less stable relationships, i.e. more changes/development over time.

Second, group differences should be indicated by different intercept values at post-treatment and follow-up, with effect sizes favoring the TG (H1.2). H1.1 was tested by comparing the information criteria of a model with freely estimated path coefficients for each group (*unequal model*) with a restricted model with fixed corresponding paths for each group (*equal model*). To test H1.2, the favored model from this first comparison was contrasted with a third model defined by fixed intercepts at post-treatment and follow-up in both groups (*fixed means model*). The model fit indices (*CFI, RMSEA, SRMR*) could not be estimated in the case of saturated models (unequal model), and sample-size adjusted Bayesian information criterion (*BIC ad.*) and Akaike information criterion (*AIC*) were used for model comparison. Favored models were indicated by lower *BIC ad.* and *AIC* values (Merkle, You, & Preacher, 2016). The model selection for non-saturated models was based on *CFI* comparisons with the fit of the unrestricted model (*CFI* equal model) being subtracted from that of the restricted model (*CFI* fixed means model). Negative *Delta CFI* values suggest the equal model is a better fit (cut-off = .002; Meade, Johnson, & Braddy, 2008). The effect sizes (Cohen's *d*) were calculated by dividing the group difference of least squares mean scores of the favored model by the pooled standard deviation for each time-point.

To inform the ZE training's implementation into autism care, additional moderator analyses were conducted for primary outcome measures in our group of interest, the TG. Autism symptomatology (SCQ), verbal age (VA), and nonverbal IQ were judged to be the most interesting for clinical practice and were therefore included as moderators in a freely estimated model (*main effects only model).* This model was compared to one including the respective moderator's interaction terms and the *GEM/KERMIT* score at baseline (*interaction model*). In the final analysis, we tested whether increases in emotional awareness or emotion regulation in the TG would predict changes in empathy/emotion recognition after training. Therefore, we calculated the *LEAS-C* change score (T2 minus T1), the *ERC-ER* change score (T2 minus T1), and the GEM/KERMIT change scores (T2 minus T1/T3 minus T2) and ran a correlation analysis. Relations between *LEAS-C/ERC-ER* change scores (T2-T1) and KERMIT/*GEM* change scores (T3-T2), were further investigated in a multiple regression analysis. All the analyses were conducted using the RStudio software version 1.3.1073 (RStudioTeam, 2015; see Table S1 for R packages and link to data repository and scripted analyzes (R markdowns). IBM SPSS statistics, version 25.0 (2017) was used for the ancillary analyses.

3. Results

Of the 184 children assessed, 82 (69 boys, 13 girls) (Fig. 2) met the inclusion criteria and were randomly assigned to ZE (n = 42 children; age range 5.3–10.8 years) or the control condition (n = 40; age range 5.5–10.6 years). There was a significantly larger proportion of females (23%; χ^2 (1) = 4.09, p = .04, φ = 0.22) in the TG than in the CG (7.5%). At baseline, the groups differed significantly in autism symptoms, as measured by SCQ, with the TG showing higher symptomatology (TG: M = 22.8, SD = 6.2; CG: M = 19.5, SD = 6.2; t (78) = 2.39, p = .019, d = 0.54). All other major demographic and clinical characteristics were comparable between groups, both for participants (Table 1) and caregivers (Table S2). Specific data on race/ethnicity were not recorded, but all the participants were native German speakers. Sample sizes varied across sites, with 37 children being enrolled at the HU, 17 in KJPP AUG, and 28 in MedUni Wien. For differences in baseline characteristics across sites and testing requirements for path analyses, see Table S1.

3.1. Attrition and missing data

Two children from the TG (5%) and four from the CG (10%) did not complete the respective training. In all cases, participants dropped out for personal reasons (e.g., death of a grandparent, separation of parents). The only significant difference between completers and non-completers was in nonverbal IQ (t (79) = −2.31, p = .024, d = −0.52), with completers (M = 104.7, SD = 18.8) scoring higher than non-completers (M = 85.0, SD = 10.2). In the TG, parent-rated data for the primary measure (GEM) was missing for 5% participants at baseline, for 26% post-treatment, and 19% at follow-up. For the CG, incomplete GEM data was given in 15% of the cases at baseline, 25% post-treatment, and 25% at follow-up. In total, 40 GEM datasets were missing across all time points (16.7%). In the TG, the KERMIT testing data was missing/invalid for 7% at baseline, 19% at post-treatment, and 24% at follow-up. At baseline, 15% of the participants of the CG had missing or invalid testing data (KERMIT), increasing to 23% at post-treatment, and 33% at follow-up (15.1% overall). Over both groups, all three time points, and all primary and secondary measures, 17.1% of data was missing/invalid (see Table S1).

3.2. Intervention monitoring: training intensity, treatment fidelity, and motivation

Training times were reported by caregivers for 59 participants (TG: n = 32; CG: n = 27; for details see Table S3). Overall, the participants trained for an average of 11.3 h (SD = 2.4) during a minimum of four weeks; the 100 min per week training requirement was met in 73% of the cases. The preferred number of training sessions per week was 4–5 for both groups, with an average length of M = 22 min in the TG (SD = 8.5) and M = 30 min in the CG (SD = 11.7). The total training times differed significantly between the groups, with higher intensity (hours) in the CG (TG: M = 9.7, SD = 2.8; CG: M = 13.2, SD = 3.6; t (57) = −4.24, p = .000). There was no significant difference in training intensity between sites (F (2,35) = 2.32, p = .114). Due to technical problems, the use of the ZE transfer module in the TG was only tracked in frequency, but not in duration. This might have caused the difference

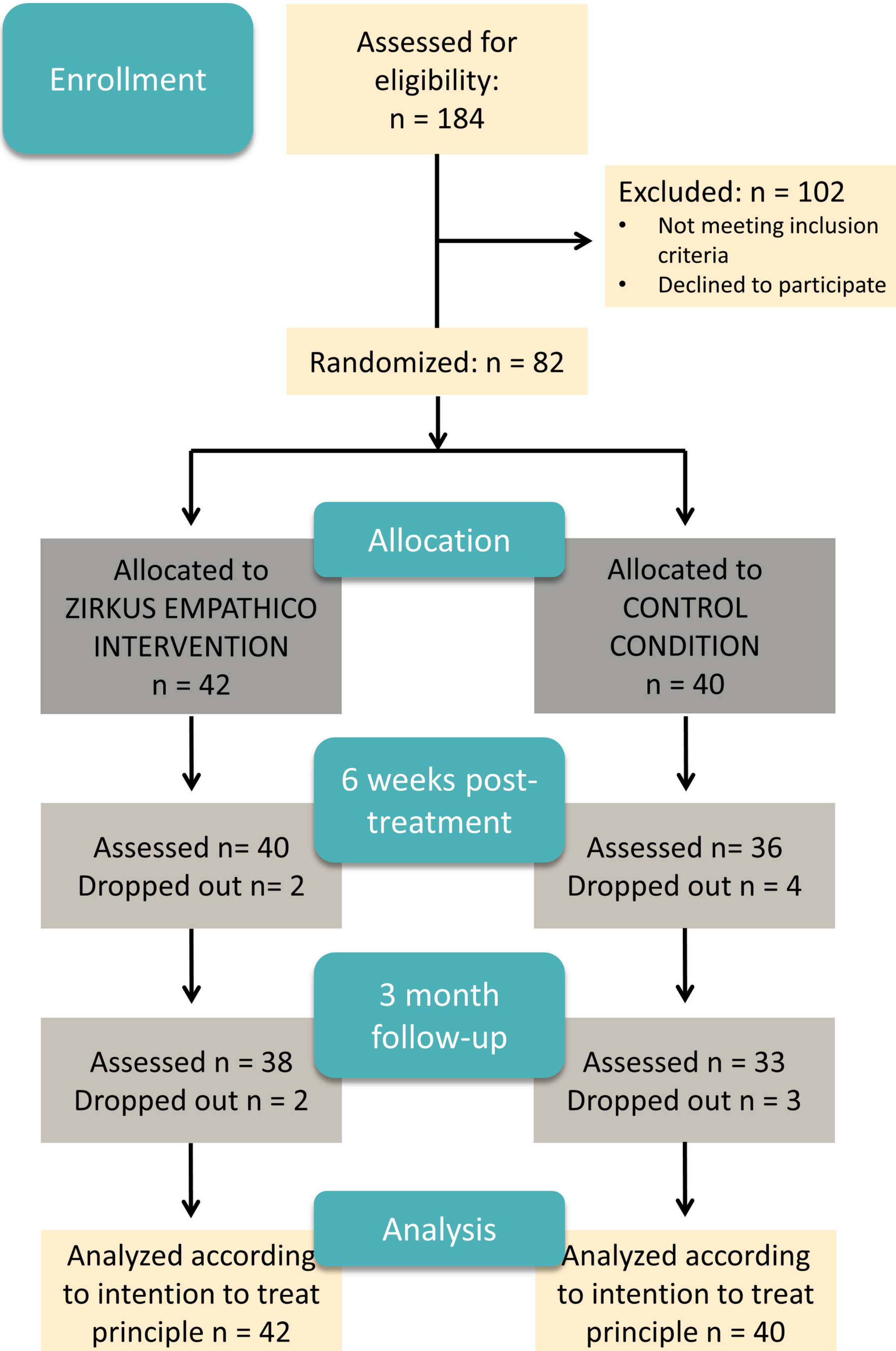


Fig. 2. Trial time flow.

Table 1
Demographic and clinical characteristics of the total sample (n = 82) by training group.

Variable	*Zirkus Empathico N = 42*			*Control Condition N = 40*			*p*
	M	SD	Range	M	SD	Range	
Age (y)	8.1	1.6	5.3–10.8	7.6	1.3	5.5–10.6	.143
CPM, Nonverbal IQ	102	19	67–135	105	19	72–135	.586
PPVT, Verbal IQ	97.6	17	66–135	103	18	73–135	.196
Verbal age (y)	7.9	2.2	4.7–13.8	7.8	2.0	5.1–13.0	.814
ADOS-G overall total (n = 60)	11.9	4.0	5–20	11.9	4.0	4–20	.997
ADOS-2 overall total (n = 13)	10.6	2.3	8–14	11.8	5.2	6–20	.600
ADI-R-short total	6.1	1.3	2–8	6.0	1.3	3–8	.678
SCQ, total score	22.8	6.2	10–35	19.5	6.2	3–31	.019
Age (y), Caregiver 1[a]	40.5	5.8	29–52	41.7	7.3	29–54	.446
AQ, Caregiver 1[a], total score	14.9	9.3	3–42	17.7	19.7	4–132	.410
TAS, Caregiver 1[a], total score	63.2	10.4	34–79	61.6	7.0	43–73	.414
	N	**%**		**N**	**%**		***p***
Males	32	76.2		37	92.5		.043
Autism diagnosis (ICD-10)							
Childhood autism	9	21.4		4	10.0		.504
Asperger syndrome	19	45.2		21	52.5		
Atypical autism	3	7.1		2	5.0		
PDD-NOS	11	26.2		13	32.5		
Co-occurring conditions							
None/Unknown	31	73.8		30	75.0		.662
ADHD/ADD	6	14.3		8	20.0		
Epilepsy	2	4,8		1	2.5		
Other	3	7.1		1.	2.5		
Pharmacological treatment							
None/unknown	35	83.3		33	82.5		.374
Central stimulants	2	4.8		1	2.5		
Antiepileptics	2	4.8		0	0.0		
Other Psychotropics	2	4.8		1	2.5		
>/ = 2 Psychotropics	1	2.4		5	12.5		
Males, Caregiver 1[a]	5	12.2		5	13.2		1.00
Level of education[b], Caregiver 1							
Lower secondary education	1	2.4		1	2.7		.780
Upper secondary education	21	51.2		16	43.2		
Academic education	19	46.3		20	54.1		
Study Center							
HU Berlin	19	45.2		18	45.0		.982
KJPP Augsburg	9	21.4		8	20.0		
UniMed Wien	14	33.3		14	35.0		

Note: ADHD/ADD = Attention Deficit Disorder, ADOS-G = Autism Diagnostic Observation Scale Generic, overall total (communication + reciprocal social interaction); ADOS-2 = Autism Diagnostic Observation Scale-2, overall total (social affect + restricted and repetitive behavior); AQ = Autism Quotient; CPM = Colored Progressive Matrices; PPVT = Peabody Picture Vocabulary Test; PDD_NOS = Pervasive developmental disorder not otherwise specified; SCQ = Social Communication Questionnaire; TAS = Toronto Alexithymia Scale.

[a] Data refers to the primary caregiver who assisted the child's training. For the second caregiver's data, see Table S2, online.

[b] The level of education was determined according to the International Standard Classification of Education (ISCED) applied to the German and Austrian school system: Lower secondary education (ISCED-Level-2); Upper secondary education (ISCED-Level-3); Academic education (ISCED-Level- 6/7).

between the automatically tracked times and the parent-recorded times in the TG, with only the latter including transfer episodes (parent-recorded: $M = 9.7$, $SD = 2.8$; tracking: $M = 7.34$, $SD = 2.4$, $t\,(29) = 4.81$, $p = .000$). However, the manually and automatically recorded times correlated significantly ($r = 0.54$, $p = .002$).

To check for treatment fidelity, the caregivers were asked whether they had been motivated to conduct the training as previously agreed upon, by using a 5-point scale (ranging from 1 'no, never', to 5 'yes, always'). According to the training protocols obtained during weekly supervision, the treatment fidelity was highly rated in both groups (TG: $M = 4.32$, $SD = 0.77$; CG: $M = 4.07$, $SD = 0.87$; $t\,(59) = 1.19$, $p = .240$). The children's motivation for engaging with the apps (volition) over the course of the training was rated as 'involved' for most participants without group differences (see Table S4 for detailed PVQ results). Retrospectively, the caregivers from both groups perceived their children as having enjoyed the training most of the time and having been mainly motivated (Treatment Satisfaction Report, Table S7a).

3.3. *Primary outcome measures*

For the *GEM* total scores, the model comparisons revealed a preference for the equal model (*AIC* = 1781.6; *BIC* = 1765.4; unequal model: *AIC* = 1785.2; *BIC* = 1766.7; Table 2) with the same regression coefficients for both groups (Table 3). Thus, no differences in the groups' developmental trajectories over time could be inferred (H1.1). Second, the analyses showed a clear preference for the model with freely estimated intercepts in both groups (H1.2): The GEM total scores differed between groups after the training, resulting in a medium effect size in favor of the TG (*difference of intercepts* between groups at T2, $DI = 17.0$; Cohen's $d = 0.71$; Fig. 3; Table S5 for means). There was no significant difference between the groups at follow-up ($DI = -4.7$; $d = -0.17$). This was supported by analyses of the accuracy of emotion recognition (*KERMIT* accuracy scores) as a second primary outcome: While the developmental trajectories and regression coefficients (Table 3) were comparable between groups (equal model: *AIC* = 1051.3, *BIC ad.* = 1035.1; unequal model: *AIC* = 1054.3, *BIC ad.* = 1035.1), the model with freely estimated intercepts revealed a between-group difference in *KERMIT* accuracy scores, with a medium effect size favoring the TG at post-treatment ($DI = 2.0$; $d = 0.50$, Fig. 3), but not at follow-up ($DI = -0.7$; $d = -0.18$).

The information criteria from the moderator analyses in the TG revealed a preference for the main effects model for both primary

Table 2
Primary and secondary outcome measures at post-treatment (week 6) and three-month follow-up: Model choice is marked. Smaller AIC/BIC values suggest a better fit, negative CFI Delta values prefer the equal model over the fixed means model (cut-off = 0.002; Meade et al., 2008).

	Model comparison I				Model comparison II		
	Unequal Model		Equal Model		Fixed Means Model		
	AIC	*BIC ad.*	*AIC*	*BIC ad.*	*AIC*	*BIC ad.*	*CFI* Delta
GEM	1785.2	1766.7	1781.6	1765.4	1791.9	1777.2	−0.08
KERMIT	1054.3	1035.7	1051.3	1035.1	1052.0	1037.3	−0.08
EKMAN	989.0	970.5	986.4	970.2	994.8	980.1	−0.17
LEAS-C	248.4	229.9	251.4	235.2	256.0	239.0	–
ERC ER	958.1	939.6	954.5	938.3	969.5	954.8	−0.12
ERC N/L	1262.6	1244.0	1256.9	1240.7	1261.9	1247.2	−0.03
SRS parent	1751.7	1733.2	1760.4	1744.2	1769.5	1752.8	–
SRS teacher	1515.8	1497.3	1523.4	1507.2	1517.9	1500.9	–
ICU	1425.5	1407.0	1420.6	1404.4	1425.7	1411.0	−0.04
Kiddy Kindl	1423.9	1405.4	1424.3	1408.1	1422.2	1407.5	–

Note: AIC = Akaike Information Criterion; BIC = Bayesian information criterion, sample size adjusted; CFI Delta = CFI Fixed Means Model – CFI Equal Model; EKMAN = Ekman & Friesen Pictures of Facial Affect Set; ERC ER = Emotion Regulation Checklist - Subscale Emotion Regulation; ERC N/L = ERC Subscale Negativity/Lability; GEM = Griffith Empathy Measure; ICU = Inventory of Callous/Unemotional Traits; KERMIT = Kids Emotion Recognition Multiple Images Tasks; LEAS-C = Level of Emotional Awareness Scale for Children, self-score; SRS = Social Responsiveness Scale.

Table 3
Standardized regression coefficients.

	Zirkus Empathico N = 42		Control Condition N = 40	
Predictor	GEM T2	GEM T3	GEM T2	GEM T3
GEM T1	.59***	–	.69***	–
GEM T2	–	.68***	–	.61***
SCQ	.08	.09	-.26**	-.22*
	KERMIT T2	KERMIT T3	KERMIT T2	KERMIT T3
KERMIT T1	.46***	–	.35***	–
KERMIT T2	–	.56***	–	.65***
SCQ	.11	-.28**	.13	-.04
	EKMAN T2	EKMAN T3	EKMAN T2	EKMAN T3
EKMAN T1	.63***	–	.44***	–
EKMAN T2	–	.23*	–	.20*
SCQ	-.02	-.21	-.20	-.28
	LEAS-C T2	LEAS-C T3	LEAS-C T2	LEAS-C T3
LEAS-C T1	.50**	–	.52***	–
LEAS-C T2	–	.74***	–	.38*
SCQ	.06	.10	.00	.14
	ERC-ER T2	ERC ER T3	ERC ER T2	ERC ER T3
ERC ER T1	.79***	–	.77***	–
ERC ER T2	–	.45***	–	.41***
SCQ	.06	-.17	-.03	-.24
	ERC-N/L T2	ERC N/L T3	ERC N/L T2	ERC N/L T3
ERC N/L T1	.75***	–	.86***	–
ERC N/L T2	–	.55***	–	.66***
SCQ	.01	-.01	-.04	.05
	SRS parent T2	SRS parent T3	SRS parent T2	SRS parent T3
SRS parent T1	.88***	–	.55***	–
SRS parent T2	–	.46*	–	.77***
SCQ	-.05	.18	.34*	.17*
	SRS teacher T2	SRS teacher T3	SRS teacher T2	SRS teacher T3
SRS teacher T1	.65***	–	.89***	–
SRS teacher T2	–	.59***	–	-.74*
SCQ	.13	.17	−0.16	.42
	ICU T2	ICU T3	ICU T2	ICU T3
ICU T1	.64***	–	.68***	–
ICU T2	–	.57***	–	.58***
SCQ	-.13	.06	.36***	.13
	Kiddy Kindl T2	Kiddy Kindl T3	Kiddy Kindl T2	Kiddy Kindl T3
Kiddy Kindl T1	.77***	–	.69***	–
Kiddy Kindl T2	–	.62***	–	.29
SCQ	-.08	-.21**	.13	.07

*p ≤ .05, **p ≤ .01, ***p ≤ .001.

measures (Table S9). Hence, the intervention effect was not impacted by any interaction between autism symptomatology, verbal age, or nonverbal IQ and the primary outcomes at baseline. After correcting for multiple comparisons, the correlation analyses (see Table S8 in the supplement) did not reveal significant correlations between the LEAS-C and the ERC-ER change scores from baseline to post-treatment (T2-T1) and the GEM and the KERMIT change scores from post-treatment to follow-up (T3-T2). Although the correlation between the LEAS change score (T2-T1) and the GEM change score (T3-T2) was only trending towards significance, the effect size was large ($r = 0.55$, $p = .061$), which is why a regression analysis was conducted. It showed that the *GEM* change score, from post-treatment to follow-up, was significantly predicted by the *LEAS-C* change score from baseline to post-treatment ($F(1,23) = 10.19$, $p = .004$, with an adjusted R^2 of 0.28).

3.4. Secondary outcome measures

When testing if training effects were expressed by differences between the groups' developmental trajectories over time (H1.1), the model comparisons revealed no differences in path coefficients between groups (preference for equal model, Table 2) for distant emotion recognition (EKMAN), emotion regulation (ERC-ER), lability/negativity (ERC-NL), and callous-unemotional traits (ICU). Thus, the two groups' developmental trajectories were comparable for these measures. The preferred unequal model indicated different developmental pathways over time between groups for emotional awareness (LEAS-C), parent-, and teacher-rated autism social symptomatology, and well-being (Kiddy-Kindl), (Table 2, and Table 3 for regression coefficients).

In the case of H1.2 (differences of intercepts between groups), an intervention effect was observed for distant emotion recognition (EKMAN: DI at T2 = 2.8, $d = 0.84$), which was not maintained at follow-up ($DI = -0.1$, $d = -0.03$). The same findings were observed for callous-unemotional traits (ICU T2: $DI = -5.3$, $d = -0.56$; T3: $DI = -0.6$, $d = -0.06$), and lability/negativity (ERC-NL T2: $DI = -3.3$, $d = -0.47$; T3: $DI = 0.2$, $d = 0.03$). However, more long-lasting changes were observed in the TG for emotion regulation (ERC-ER T2: $DI = 2.3$, $d = 0.70$; T3: $DI = 1.2$, $d = 0.33$) and the awareness of their own emotions (LEAS-C T2: $DI = 0.3$, $d = 0.67$; T3: $DI = 0.1$, $d = 0.24$). Importantly, parents and teachers reported a reduction in autism social symptomatology (SRS) directly after training (parents: $DI = -18.2$; $d = -0.69$; teacher: $DI = -7.0$; $d = -0.28$), and at follow-up (parents: $DI = -7.1$; $d = -0.28$; teacher: $DI = 13.7$; $d = -0.53$). Model comparisons with a preference for the unequal model and a rejection of the fixed means model indicate a positive effect on the children's wellbeing (Kiddy Kindl) from the ZE training. However, the effect sizes were too small to be of significance

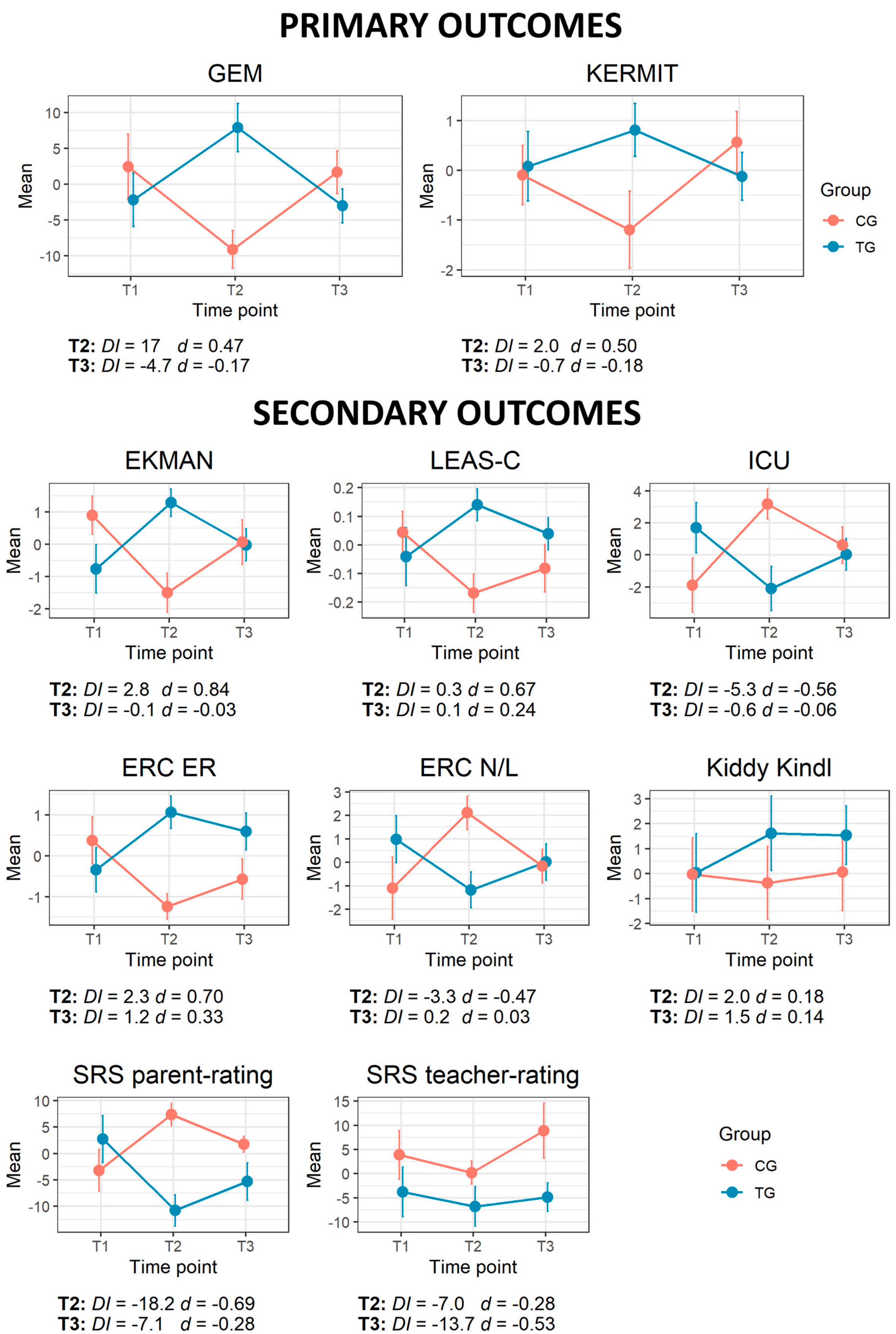


Fig. 3. Graphs showing group means at baseline (T1) and estimated means (unstandardized intercepts) at post-treatment (T2), and follow-up (T3) for primary and secondary outcomes. The differences in intercepts (*DI*) and effect sizes (Cohen's *d*) at T2 and T3 are presented for each outcome.

(Cohen's d < 2; Cohen, 1988; T1: $DI = 2.0$, $d = 0.18$; T2: $DI = 1.5$; $d = 0.14$).

3.5. Treatment satisfaction and feasibility

As reported in the Treatment Satisfaction Report (online report/ interview), treatment satisfaction and acceptance (e.g., enjoyment, motivation to play) were generally high after the intervention in caregivers (TG: n = 34; CG: n = 27) and children (TG: n = 34, CG: n = 22), and did not differ between the groups, except that the caregivers' enjoyment was significantly higher in the TG. Overall, the feasibility of the training, operationalized by compatibility with family life, training-induced stress, quality of supervision, and understanding of training content, was good. Adverse events were not accessed systematically, but caregivers of both groups did not report negative training effects within the weekly training supervision. In the caregiver report at T2, the children were reported to show low negative emotionality (e.g., stress, frustration) during training and caregivers rated for themselves having been only minimally stressed by the intervention**.** For a detailed report and the results of the children's interviews, see Table S7a/b.

3.6. Treatment goal achievement

The caregivers' treatment satisfaction reports for the ZE TG indicated that the majority of participants showed enhanced interest in, and recognition of emotions, improvements in dealing with, and awareness of their own emotions, and adequate reactions towards other people's emotions. Sixty percent of the caregivers in the TG experienced their child being more sociable than before treatment. The TG and CG differed significantly when caregivers were asked whether their child had changed its behavior, and whether their emotional approach, relationship, and/or interaction/communication with their child had improved during the training, with higher values in the TG (see Table S8a/b for details and the results of the children's interviews). In addition, it was possible to interview 30 parents (71%) in the ZE TG (n = 42), post-treatment, concerning treatment goal achievement (GAS; for details Table S9). Parents reported achievement (27%; GAS score = 0) or over-achievement (39%; GAS score = 1 or 2) for n = 60 predefined goals. A subtle change in behavior towards the goal (GAS score = −1) was observed in 28% of the goals, while goals were not reached in 6% (GAS score = −2).

4. Discussion

Although serious games have previously been identified as promising ways of teaching socio-emotional skills to children on the AS, a lack of RCTs investigating the maintenance and transfer of skills to everyday behavior limits the external validity of the findings (Berggren et al., 2017; Grynszpan et al., 2014; Kouo & Egel, 2016). Beyond that, the different facets of empathy have also not yet been adequately addressed in conjunction with emotion recognition. The current RCT addresses these points by testing a parent-assisted serious game about socio-emotional competencies with a follow-up assessment, and quantitative and qualitative generalization measures. In contrast to most previous studies, an active control group was integrated into the study to account for maturation effects and the effects of enhanced parent-child interaction and media use. Furthermore, using path analyses allowed differences in the groups' developmental trajectories over time to be modeled and unbiased changes assessed.

While short-term intervention effects on general empathy and emotion recognition, as primary outcomes, could only be demonstrated directly after training, the training seemed to be effective in the short-, *and* mid-term for skills related to the processing of the participants' *own* emotions, i.e., emotional awareness and regulation. Furthermore, the reduction of autism social symptomatology in parent and teacher reports, the positive results of the parents' and children's treatment satisfaction reports, and the goal attainment scaling are indicative of a transfer of socio-emotional skills into different, real-life settings (family/ school/kindergarten). Given the high training acceptance, as indicated by parent report, low drop-out rates, and satisfactory treatment fidelity, we conclude that ZE appears to be a promising intervention for children on the AS without severe intellectual impairment. Several interesting observations are relevant to practical implementation and future program development and research.

4.1. Empathy and emotion recognition

For the primary outcomes, improvements could be demonstrated in (1) parent-rated empathy and (2) emotion recognition abilities in the TG after six weeks of training, resulting in medium effect sizes (Cohen, 1988). Neither effect was moderated by autism symptomatology (SCQ) or verbal age. These results accord well with previous research on serious games for fostering socio-emotional competencies in children on the AS (Grynszpan et al., 2014). However, no differences between the ZE training and control groups were present, for either construct, three months after the training.

This pattern was replicated by the EKMAN emotion recognition test as a secondary measure: The large post-treatment intervention effect suggests a distant generalization effect to untrained stimuli, but the effect was not maintained three months later. These findings are in contrast to Thomeer and colleagues' (2015) evaluation of the serious game *Mind Reading*, which showed a medium treatment effect for emotion recognition at a five-week follow-up assessment in children on the AS and an IQ > 70, in comparison to waitlist controls. The different results may be because *Mind Reading* focuses exclusively on emotion recognition, with richer content targeting simple and complex emotion recognition in facial and bodily expressions through multiple programs and games, and a higher dose of training (12 weeks).

In contrast, ZE focuses solely on unambiguous basic emotions without the inclusion of progressive levels of difficulty and individualization (e.g., matching the task difficulty to the child's abilities), which may have resulted in insufficient learning trials and early ceiling performance, as outlined by Whyte et al. (2015). Furthermore, the *intrinsic* motivation to acquire emotion recognition skills was not addressed, either during or after the intervention (e.g., through developing individual and significant goals with the children). Thus, the ZE training might have facilitated attentiveness toward facial emotions during the training, while lacking motivation and refreshment of skills after the end of the intervention, which might have led to reduced maintenance. Finally, the recognition of other people's emotions seems to have been difficult for caregivers and children to generalize: As parents stated during training supervision, naturally occurring emotional expressions are more transient and mixed than those provided in the video content (Riediger, Voelkle, Ebner, & Lindenberger, 2011), and thus not easy to capture.

4.2. Changes in socio-emotional traits and autism social symptomatology

Interestingly, callous-unemotional traits that have been reported to be associated with deficits in fear recognition in typically and atypically developing samples (Leno et al., 2015), only improved in the short-term, similarly to emotion recognition skills. The same was true for lability-negativity (lack of flexibility, anger dysregulation, mood lability), which was reduced only over the training period. We might speculate that more specific content (e.g., different and multimodal fear stimuli, emotion regulation skills for dealing with frustration and anger) and higher intensity training would have resulted in more long-lasting changes.

In contrast to emotion recognition/empathy, our findings indicate mid-term changes in emotional awareness, emotion regulation (measured by expression of emotions, empathy, and constructive emotional self-awareness), and autism social symptomatology, as demonstrated by medium effect sizes after training and, albeit smaller, at follow-up. For emotional awareness and autism social symptomatology, the differences in the groups' developmental pathways over time further emphasized the changes observed in the TG in comparison to the active controls. Furthermore, the increased awareness of their own emotions during the intervention predicted gains in empathy three months after the intervention, which might suggest that the process of differentiating own emotions contributes to empathy (compare Bird & Viding, 2014; Cook, Brewer, Shah, & Bird, 2013; Moriguchi et al., 2006; Moriguchi et al., 2007).

Interestingly, changes in emotional awareness were not associated with improvements in emotion recognition after training, and changes in emotion recognition within training were not related to changes in empathy after training. These findings are in contrast with the results of other studies, for example, Israelashvili et al. (2019; 2020). This might be due to differences in methodology (e.g., tasks targeting complex versus basic emotions, non-autistic adult samples versus children on the AS). To conclude, future studies should explore whether and how tasks that strengthen the awareness of own emotions/emotion recognition indeed affect different facets of empathy as suggested by models of empathy (e.g., Bird & Viding, 2014; Shamay-Tsoory, 2011).

The positive results for social functioning (here: SRS) and emotion regulation accord well with other studies, e.g., on the effectiveness of the *Secret Agent Society* (Beaumont & Sofronoff, 2008; Einfeld et al., 2018; Sofronoff et al., 2017), *FaceSay™* (Hopkins et al., 2011; Rice et al., 2015), and *Emotiplay* (Fridenson-Hayo et al., 2017). Nevertheless, the difference between the more pronounced changes in the participants' processing of their own emotions and social functioning versus only short-term training effects in processing others' emotions warrants further examination. Interestingly, Beaumont and Sofronoff (2008) reported similar results, with improvements in social functioning and emotion regulation, but a lack of training effect on behaviorally assessed emotion recognition, which was explained by ceiling and practice effects in the behavioral measure used.

Another possible explanation of these, and our, results may be that parents perceived interventions that targeted their child's own emotions as being more relevant to their daily family lives and therefore focused specifically on those interventions when working with their children. Indeed, research shows that behavioral and emotional dysfunction, such as anger dysregulation and temper tantrums, interfere greatly with family life and are associated with parental stress, anxiety, and depression, while reduced social competencies have a lesser impact (Davis & Neece, 2017; Firth & Dryer, 2013). This psychological strain may have resulted in more parental effort and motivation being focused on fostering emotional awareness and regulation during and after training, in comparison to repeatedly focusing on other people's emotional expressions. In addition, the children's social environments may have provided reinforcement, especially for enhanced emotion regulation and communication skills, resulting in better maintenance. Indirect evidence for this hypothesis comes from parents' statements during training supervision and the training goals defined by parents in the goal attainment scaling: In almost half of the cases, the parents wished to improve their children's handling of their own emotions (47%). Finally, because the ZE modules focusing on own emotions (I, IV) used videotaped emotion-eliciting contexts, they were more immersive and contextualized than module II, which presented isolated facial expressions. According to Whyte et al. (2015), more extensive learning contexts should support the learning of difficult behaviors and skills, as the motivation to learn them is intrinsically enhanced in the children. However, due to the lack of further studies which investigate the maintenance effects of computer-based interventions on emotion recognition (Berggren et al., 2017), empathy and related socio-emotional skills, there is a need for additional RCTs with rigorous methodology (e.g., active controls, long-term assessment, distant generalization measures).

4.3. Implications

In view of the generally high acceptance, feasibility, treatment satisfaction, and good ecological transfer of serious game effects to social abilities in daily-life situations (e.g., Beaumont & Sofronoff, 2008; Thomeer et al., 2015), our study's results are promising regarding practical implementation of the serious game *Zirkus Empathico* into current autism interventions, e.g., as a complement to specialized emotional and social skills training, such as that recommended by Berggren et al. (2017). As reported by the parents of our study participants, we see an especially high potential for ZE to serve as a communication context for sharing emotional experiences with peers and family members. In this function, the app facilitates the parents' involvement in the training of socio-emotional skills, which has already been shown to be important for therapy transfer and outcome (McConachie & Diggle, 2007; Sofronoff, Attwood, & Hinton, 2005; Sofronoff et al., 2017). In addition, and considering the "double empathy problem" (Milton, 2012), which posits that non-autistic individuals generally have just as much difficulty in understanding the autistic mind as vice versa, ZE can foster the mutual understanding of emotion processing and thus, can help non-autistic individuals (e.g., caregivers, peers) to empathize more with children on the AS.

While designing the ZE intervention, we respected the importance of embedding serious games into real-life settings in order to enhance generalization, as it has been emphasized by other authors (see Kouo & Egel, 2016). However, as our findings for emotion recognition and empathy are limited, it might be fruitful to give parents more intensive training in providing structured real-life practices, and to incorporate more motivating transfer tasks/materials that are linked to the ZE storyline, for the children (compare Beaumont & Sofronoff, 2008; Whyte et al., 2015). Future research on ZE should investigate whether longer training periods under structured, professional supervision would result in more pronounced treatment effects and whether the content is more appropriate and effective for younger age groups. Beyond that, and to test the effectiveness of ZE more comprehensively, future studies should investigate if our results apply in other samples with training-assisting caretakers having lower educational or different ethical backgrounds, or less time. Currently, our results do apply for predominantly female caretakers with mostly Caucasian ethnicity, medium to high educational background, who were able to investigate up to 100 min per week in their child's training. Furthermore, the feasibility of using ZE in routine care, as well as for fostering socio-emotional competencies in associated conditions (e.g., ADHD, anxiety disorders), should be ascertained.

Finally, when considering future serious game development, implementation, and research, it might be fruitful to compare the different computer-based interventions with each other to investigate which specific training content, elements, and mechanisms (e.g., various tasks/materials, individualized/adaptive difficulty, compare Whyte et al., 2015) are most suitable for affecting targeted outcomes for different contexts (e.g., settings, assisting facilitators, target groups). Specifically, factors that support the successful generalization and maintenance of social skills (compare Gunning et al., 2019), should be examined so that they can be carefully incorporated into future serious game design.

4.4. Limitations

There are several limitations to the study, which are in part due to the assessment having taken place in the children's natural setting. First, because of structural limitations (e.g. time restrictions), real-life transfer was assessed using parent and teacher reports rather than direct and more objective, behavioral observations. Furthermore, as is common in psychotherapy studies, the observers could not be blinded regarding treatment allocation and therefore, the positive effects could have resulted, at least in part, from observer biases. However, given that the behavioral results from the objective tests support the parents' and teachers' subjective observations, it is unlikely that the effects are solely attributable to observer bias. An active control group was included to account for the potential effects of enhanced parent-child interaction, media use, and professional supervision, which could potentially have driven the results. Although the children were allowed to receive other treatments in parallel with the serious game, no socio-emotional content was allowed, so that a confounding of our results is unlikely.

A future study on parent-assisted serious games should investigate the caregivers' and children's behavior during training (e.g. by filming and analyzing training sessions) to allow insights into learning and other moderators that could potentially drive training effects. Furthermore, adverse events should be assessed more systematically to ensure safe training environment and a more complete picture of the effects of the training. While behavioral paradigms allow measuring direct effects on the children's behavior, they should be piloted beforehand to ensure a sufficient understanding of the task requirements, which was not the case for the MET-J/K in our sample. Finally, about 17% of the data overall participants and time points were missing because the outcome measures depended largely on the caretakers' compliance. This problem could be handled partly by using a full information maximum likelihood estimation method in statistical analyses. Nevertheless, this was not possible for evaluation of the treatment goal achievement (GAS) within the treatment group. To conclude, future studies on serious games should include more direct, behavioral, or observer-based assessments of the targeted competencies and rely less on parental judgment.

5. Conclusions

This study has shown that the parent-assisted serious game *Zirkus Empathico* has some potential for training socio-emotional skills in children on the autism spectrum. Future research is needed to investigate the specific mechanisms, which account for the transfer and maintenance of emotion recognition and empathy. However, the broad approach taken to the own and other people's socio-emotional functioning and the intervention's naturalistic approach seemed to be effective in improving emotional awareness and emotion regulation, as well as general autism symptomatology in the mid-term.

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CRediT authorship contribution statement

Simone Kirst: Conceptualization, Software, Methodology, Funding acquisition, Project administration, Formal analysis, Visualization, Writing – original draft, preparation. **Robert Diehm:** Investigation, Data curation, Writing – review & editing. **Katharina Bögl:** Methodology, Formal analysis, Writing – review & editing. **Sabine Wilde-Etzold:** Investigation. **Christiane Bach:** Methodology, Funding acquisition. **Michele Noterdaeme:** Resources, Supervision. **Luise Poustka:** Resources, Methodology, Supervision. **Matthias Ziegler:** Methodology, Formal analysis, Supervision, Writing – review & editing. **Dziobek Isabel:** Conceptualization, Supervision, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

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Prof. Dr. Luise Poustka was with the Department of Child and Adolescent Medicine of the Medical University of Vienna, Vienna, Austria. She is now with the Department of Child and Adolescent Psychiatry and Psychotherapy, University Medicine, Georg-August Universität Göttingen, Göttingen, Germany.